



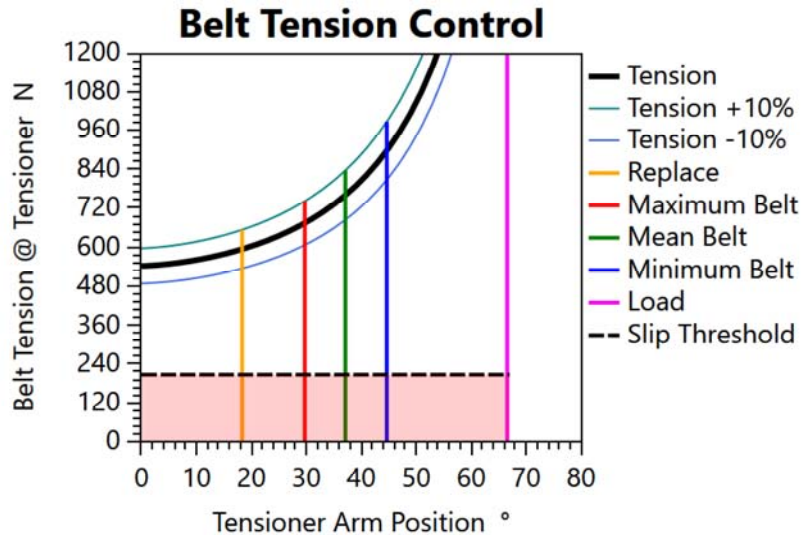
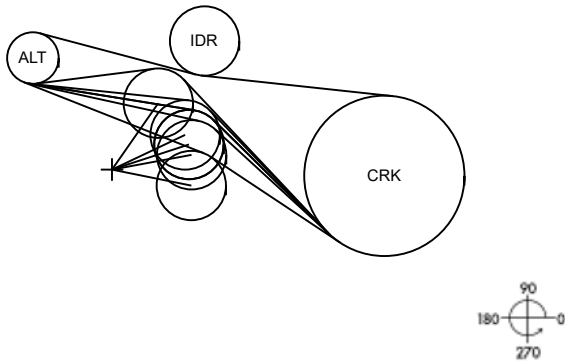
AG0810 BMC Otomotiv TTA-6x6
250Amp ALT Drive
T38519 (29.5Nm): CCW@-303/7
10PK1215HD MT610 Polyester
Vers-8; 16Sep2021

Belt Data, DB Ver. = 2.13, Belt Rev. = 7/14/2016

Belt Code	MT610
# of Ribs / Cord Material	10 / polyester
Effective Belt Length mm (ISO 9981)	1214

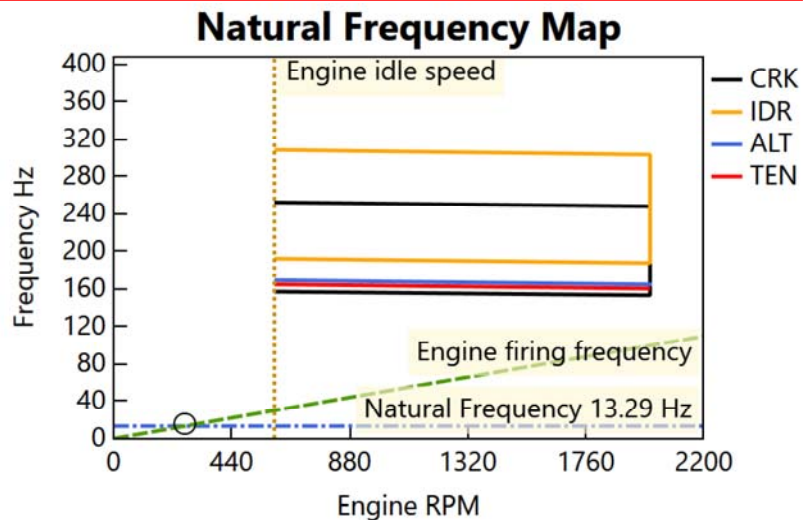
Tensioner Data

Type	Automatic
Design Tension N	759 (171 lbf)



Belt Life B10

Duty Cycle DCn-th %	Av. Vehicle Speed km/hr	PK-2_2a-MT3 hours
95	1	2559



Peak Tension & Hubload

	Engine RPM	Accel. RPM/s	Tension N	Hubload N	Direction /Wrap °
CRK	600	1000	2530	3160.4	165 / 218
IDR	600	1000	2527	384.5	80 / 9
ALT	600	1000	763	3287.4	346 / 186
TEN	600	1000	759	462.9	244 / 35

Note: Peak Loads are defined as the highest loads (tension, shear stress and hubload) calculated for all peak load & accelerations conditions. (Not from system vibration calculation)

Permissible Grooved Pulley

Axial Offset Allowance

For 0.90° Belt Entry Angle

Adjacent Grooved Pulleys	Flat Pulley Angular Misalignment °	Axial Offset Allowance mm	Grooved Pulley Axial Offset %	Attributed to Flat Pulley Angular Misalignment %
ALT->CRK	IDR (0.33)	3.46	76	24
CRK->ALT	TEN (1.00)	1.4	10	90

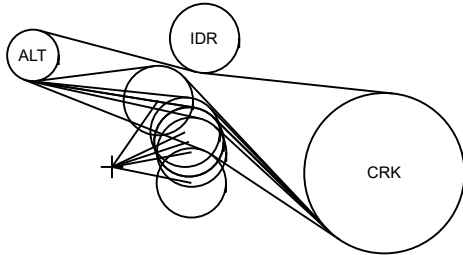


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250Amp ALT Drive
T38519 (29.5Nm): CCW@-303/7
10PK1215HD MT610 Polyester
Vers-8; 16Sep2021

Layout Data mm

Layout # 0

	X	Y	Flat	Pitch	Effective	DOB
CRK	0.00	0.00		178.40	176.00	176.99
IDR	-200.00	150.00	75.00	77.20	79.60	
ALT	-391.00	131.00		57.40	55.00	55.99
TEN	-217.41	34.81	75.00	77.20	79.60	



Belt Data, DB Ver. = 2.13, Belt Rev. = 7/14/2016

Belt Code	MT610
# of Ribs / Cord Material	10 / polyester
Stretch and Wear Allowance % of Length	0.6
Belt Length Tolerance (+/-) mm	5

Tensioner Data

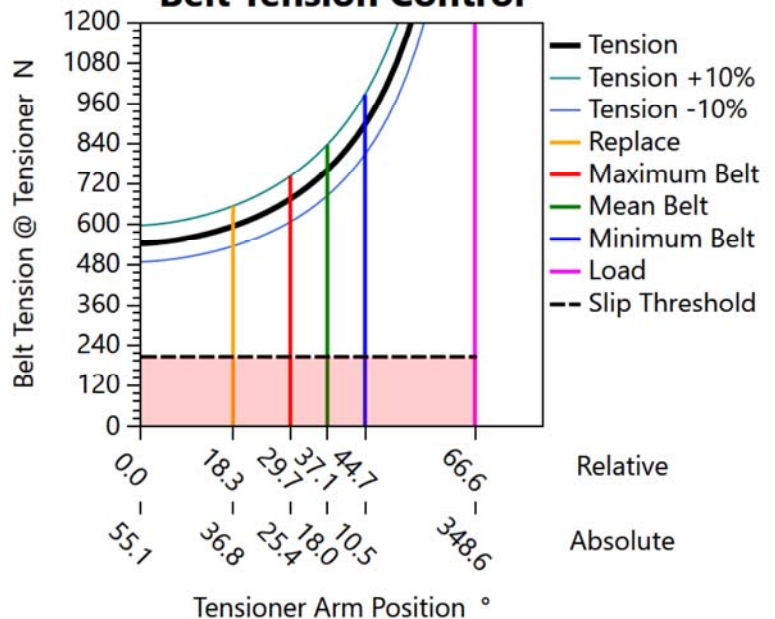
Type	Automatic
Design Tension N	759 (171 lbf)
Pivot Point {X, Y Coordinates} mm	-303.00, 7.00
Arm Length mm	90.0
Spring Mean Load Nm	29.48
Spring Rate Nm/deg	0.483

RESULTS

Belt Drive System Geometry

	Span Length mm	Wrap Angle °	Speed Ratio (Ref. Engine)
CRK	214.9	218.5	1.000
IDR	179.8	8.7	2.311
ALT	186.7	185.7	3.108
TEN	179.3	35.4	2.311

Belt Tension Control





AG0810 BMC Otomotiv TTA-6x6

250Amp ALT Drive

T38519 (29.5Nm): CCW@-303/7

10PK1215HD MT610 Polyester

Vers-8; 16Sep2021

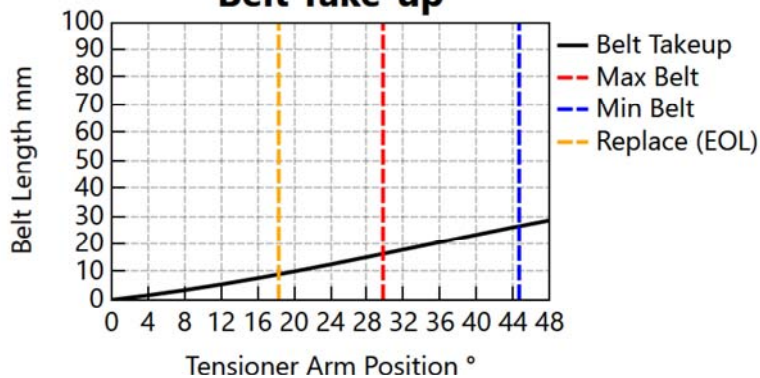
Tensioner Geometry

Belt Takeup / Tensioner Arm Ratio mm/°

0.677

Position Description	Belt-Drive Length					
	Free Arm	Replace	Max	Mean	Min	Load
Arm Position °	55.1	36.8	25.4	18.0	10.5	348.6
Idler X Coordinate mm	-251.6	-231.0	-221.7	-217.4	-214.5	-214.8
Idler Y Coordinate mm	80.8	60.9	45.6	34.8	23.4	-10.8
Hubload to Arm Angle °	15.1	29.1	38.6	45.1	52.1	73.9
Hubload Direction °	250.2	245.9	244.0	243.1	242.5	242.5
Hubload N	492.0	465.4	461.2	461.8	466.4	505.3
Tension N	542.2	593.8	674.0	759.2	896.5	2404.6
Idler Wrap Angle °	54.0	46.1	40.0	35.4	30.2	12.1
Effective Drive Length mm	1237.3	1228.2	1220.9	1215.9	1210.9	1199.8
Required Eff. Belt Length mm	1235.7	1226.6	1219.3	1214.3	1209.3	1198.2

Belt Take-up

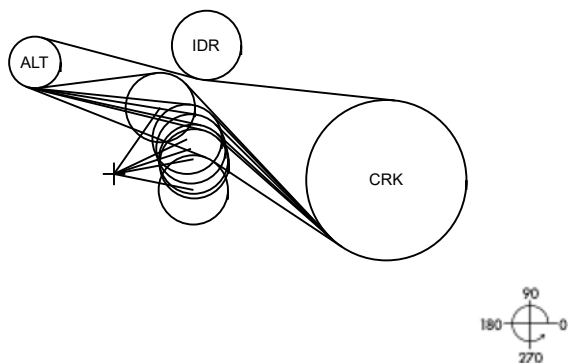




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250Amp ALT Drive
T38519 (29.5Nm): CCW@-303/7
10PK1215HD MT610 Polyester
Vers-8; 16Sep2021

Alignment Data °

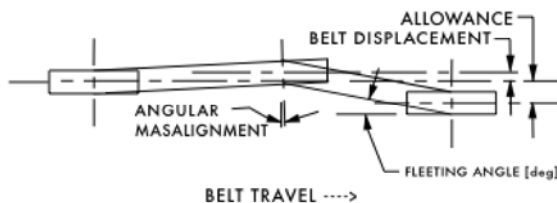
Fleeting Angle	0.90
IDR Angular Misalignment	0.33
TEN Angular Misalignment	1.00



(GROOVED - GROOVED) PULLEYS



(GROOVED - FLAT - GROOVED) PULLEYS



RESULTS

Permissible Grooved Pulley

Axial Offset Allowance

For 0.90° Belt Entry Angle

Adjacent Grooved Pulleys	Flat Pulley Angular Misalignment °	Axial Offset Allowance mm	Grooved Pulley Axial Offset %	Attributed to Flat Pulley Angular Misalignment %
ALT->CRK	IDR (0.33)	3.46	76	24
CRK->ALT	TEN (1.00)	1.4	10	90

Backside Pulley Alignment Sensitivity

Backside Pulley	Angular Misalignment	Most Sensitive Direction	Resulting Fleeting Angle	Suggestions
2: IDR	0.330	76.65	0.22	AB C E
4: TEN	1.000	70.71	0.81	AB C E

Suggestion List for Backside Pulley Alignment Sensitivity

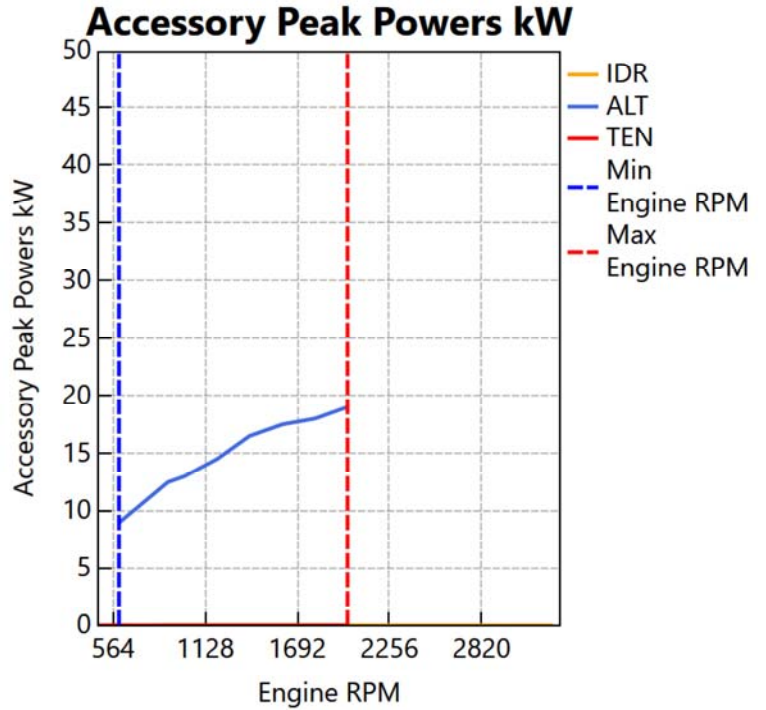
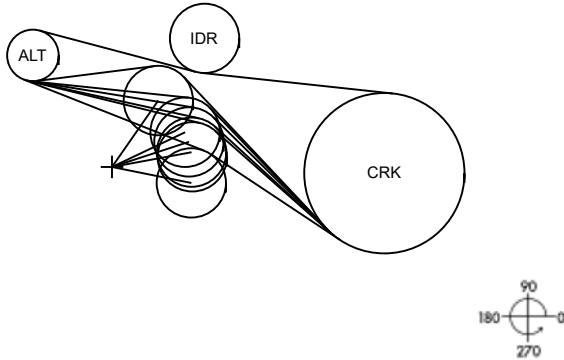
- A] Increase EXIT span length
- B] Decrease ENTRANCE span length
- C] Increase both ENTRANCE/EXIT span lengths
- D] Increase wrap angle
- E] Decrease wrap angle

Note: Most Sensitive Direction for Grooved pulleys is parallel to the belt entry span.



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Vers-8; 16Sep2021

Design Tension N	759 (171 [lbf])
Engine Acceleration RPM/s	1000
Engine Deceleration RPM/s	1000



Accessory Data

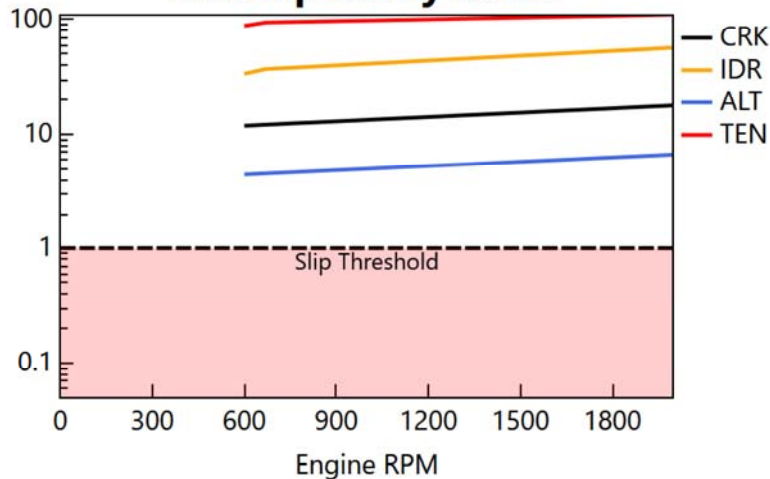
	File Name - if used	Moment of Inertia kg m ²
IDR	IDR.cmp	0.0002
ALT	AG810-250Amp-ALT.cmp	0.0140
TEN	TEN.cmp	0.0004

RESULTS

Belt Slip Sensitivity

Pulley	Most Critical Load Condition					Required Increase Tension or Wrap Angle	
	Acceler. RPM/s	Load Combination %				N	°
CRK	1000	100					
IDR	1000	10					
ALT	1000	100					
TEN	1000	100					

Belt Slip Safety Factor





AG0810 BMC Otomotiv TTA-6x6

250Amp ALT Drive

T38519 (29.5Nm): CCW@-303/7

10PK1215HD MT610 Polyester

Vers-8; 16Sep2021

Design Tension N

759 (171 [lbf])

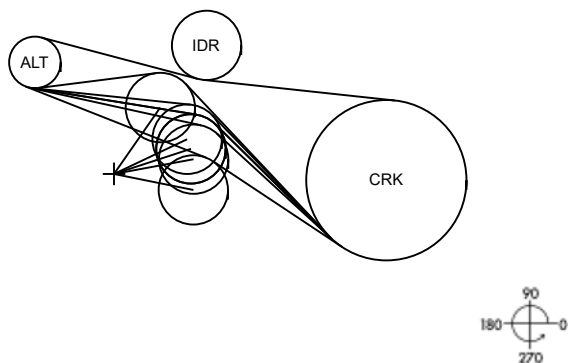
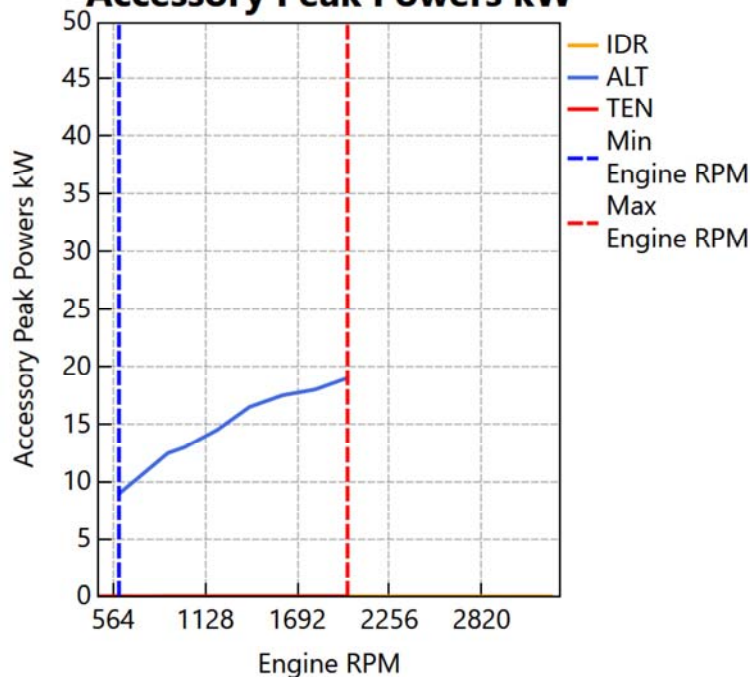
Engine Acceleration RPM/s

1000

Engine Deceleration RPM/s

1000

Accessory Peak Powers kW



RESULTS

Peak Tension & Hubload

	Engine RPM	Accel. RPM/s	Tension N	Hubload N	Direction /Wrap °	Shear Stress N/mm-rib @ RPM
CRK	600	1000	2530	3160.4	165 / 218	0.535 @ 600
IDR	600	1000	2527	384.5	80 / 9	N/A
ALT	600	1000	763	3287.4	346 / 186	2.066 @ 600
TEN	600	1000	759	462.9	244 / 35	N/A

Note: Peak Loads are defined as the highest loads (tension, shear stress and hubload) calculated for all peak load & accelerations conditions. (Not from system vibration calculation)



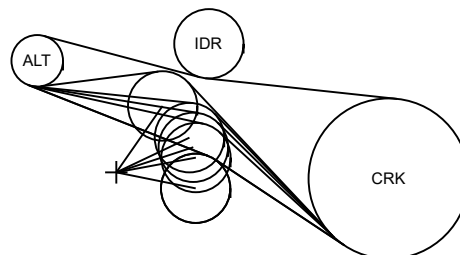
AG0810 BMC Otomotiv TTA-6x6

250Amp ALT Drive

T38519 (29.5Nm): CCW@-303/7

10PK1215HD MT610 Polyester

Vers-8; 16Sep2021



Load Conditions kW for DC 95%

%	Eng. RPM	V, km/hr	T, °C	CRK	IDR	ALT	TEN
27.00	600	1	90	9.02	0.01	9.00	0.01
0.00	900	1	90	12.52	0.01	12.50	0.01
10.00	1000	1	90	13.02	0.01	13.00	0.01
13.00	1200	1	90	14.52	0.01	14.50	0.01
0.00	1400	1	90	16.52	0.01	16.50	0.01
18.00	1500	1	90	17.02	0.01	17.00	0.01
0.00	1600	1	90	17.52	0.01	17.50	0.01
19.00	1800	1	90	18.02	0.01	18.00	0.01
0.00	1900	1	90	18.52	0.01	18.50	0.01
13.00	2000	1	90	19.02	0.01	19.00	0.01

Mean Hubloads N for DC 95%

%	Eng. RPM	CRK	IDR	ALT	TEN
27.00	600	3001	360	3125	462
0.00	900	2882	342	3005	462
10.00	1000	2788	327	2910	462
13.00	1200	2691	312	2811	462
0.00	1400	2659	307	2779	462
18.00	1500	2612	300	2730	462
0.00	1600	2570	294	2688	462
19.00	1800	2471	278	2588	462
0.00	1900	2443	274	2559	462
13.00	2000	2418	270	2534	462

Mean Tensions N for DC 95%

%	Eng. RPM	CRK	IDR	ALT	TEN
27.00	600	2369	2367	761	759
0.00	900	2248	2247	760	759
10.00	1000	2153	2152	760	759
13.00	1200	2055	2054	760	759
0.00	1400	2022	2022	760	759
18.00	1500	1974	1973	760	759
0.00	1600	1931	1931	760	759
19.00	1800	1831	1830	760	759
0.00	1900	1803	1802	760	759
13.00	2000	1777	1777	760	759



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250Amp ALT Drive
T38519 (29.5Nm): CCW@-303/7
10PK1215HD MT610 Polyester
Vers-8; 16Sep2021

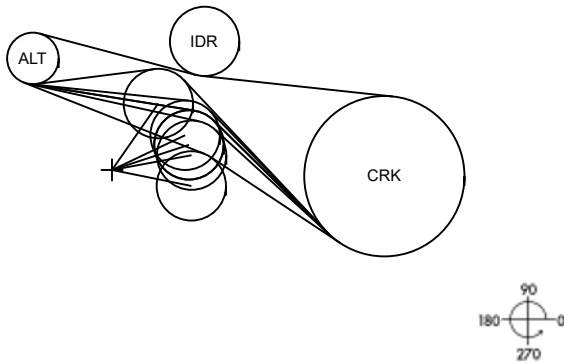
Load Conditions [kW]

Belt Data, DB Ver. = 2.13, Belt Rev. = 7/14/2016

Belt Code	MT610
# of Ribs / Cord Material	10 / polyester

Tensioner Data

Type	Automatic
Design Tension N	759 (171 lbf)





AG0810 BMC Otomotiv TTA-6x6

250Amp ALT Drive

T38519 (29.5Nm): CCW@-303/7

10PK1215HD MT610 Polyester

Vers-8; 16Sep2021

RESULTS

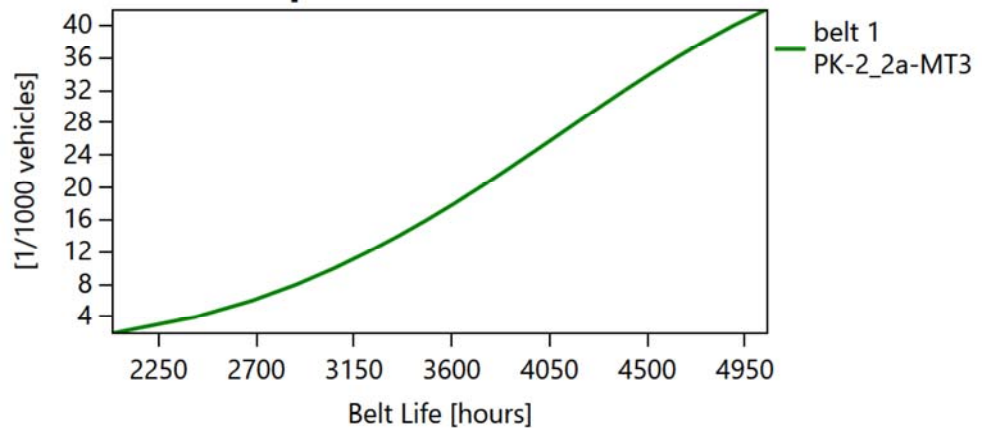
Pulley Contributions to Belt Rib Fatigue %

	CRK	IDR	ALT	TEN
PK-2_2a-MT3	0.6	16.4	66.7	16.4

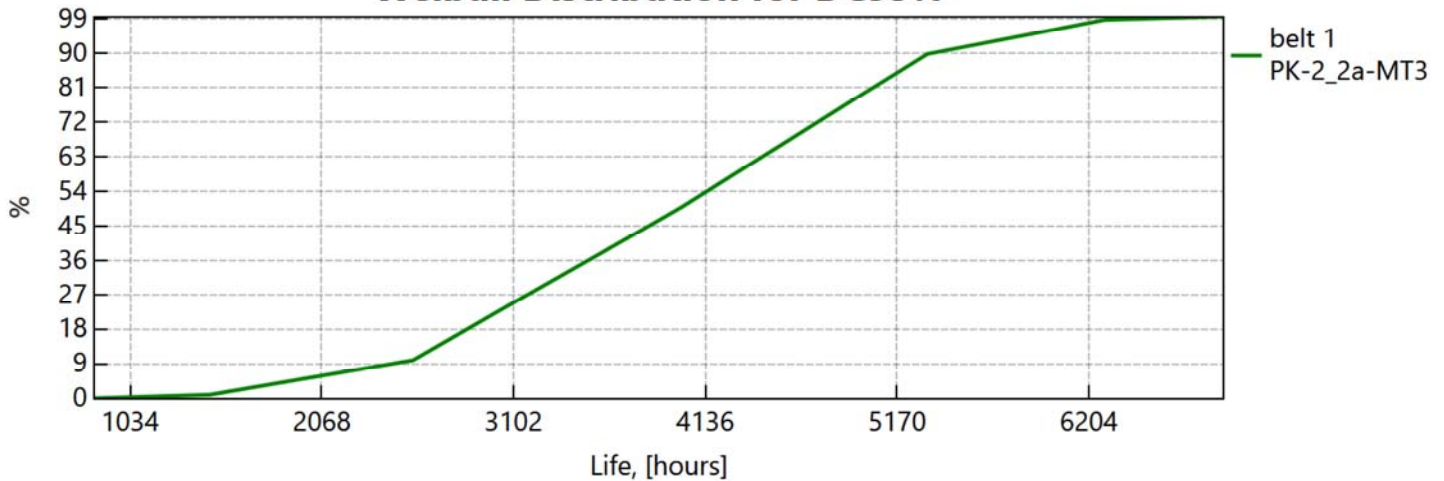
Belt Life B10

Duty Cycle DCn-th %	Av. Vehicle Speed km/hr	PK-2_2a-MT3 hours
95	1	2559

Belt Incidents per Thousand Vehicles



Weibull Distribution for DC95%



Load Conditions kW for DC 95%

%	Eng. RPM	V, km/hr	T, °C	CRK	IDR	ALT	TEN
27.00	600	1	90	9.02	0.01	9.00	0.01
0.00	900	1	90	12.52	0.01	12.50	0.01
10.00	1000	1	90	13.02	0.01	13.00	0.01
13.00	1200	1	90	14.52	0.01	14.50	0.01
0.00	1400	1	90	16.52	0.01	16.50	0.01
18.00	1500	1	90	17.02	0.01	17.00	0.01
0.00	1600	1	90	17.52	0.01	17.50	0.01
19.00	1800	1	90	18.02	0.01	18.00	0.01
0.00	1900	1	90	18.52	0.01	18.50	0.01
13.00	2000	1	90	19.02	0.01	19.00	0.01

Load Contribution to Belt Rib Fatigue for DC 95%

%	VbIt m/s	CRK Tt N	CRK Ts N	% PK-2_2a-MT3
27.00	5.6	2369	759	18
0.00	8.4	2248	759	0
10.00	9.3	2153	759	9
13.00	11.2	2055	759	13
0.00	13.1	2022	759	0
18.00	14.0	1974	759	21
0.00	14.9	1931	759	0
19.00	16.8	1831	759	23
0.00	17.7	1803	759	0
13.00	18.7	1777	759	16



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T38519 (29.5Nm): CCW@-303/7
10PK1215HD MT610 Polyester
Vers-8; 16Sep2021

Belt Data, DB Ver. = 2.13, Belt Rev. = 7/14/2016

Belt Code	MT610
# of Ribs / Cord Material	10 / polyester

Tensioner Data

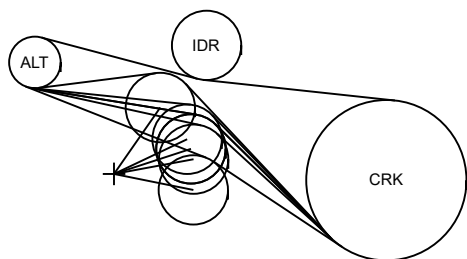
Type	Automatic
Design Tension N	759 (171 lbf)
Inertia of the Tensioner Arm Kg-m ²	0.0040
Tensioner Pulley Mass kg	0.80

Engine Data

# of Cylinders	6
Crankshaft Moment of Inertia kg m ²	0.7000

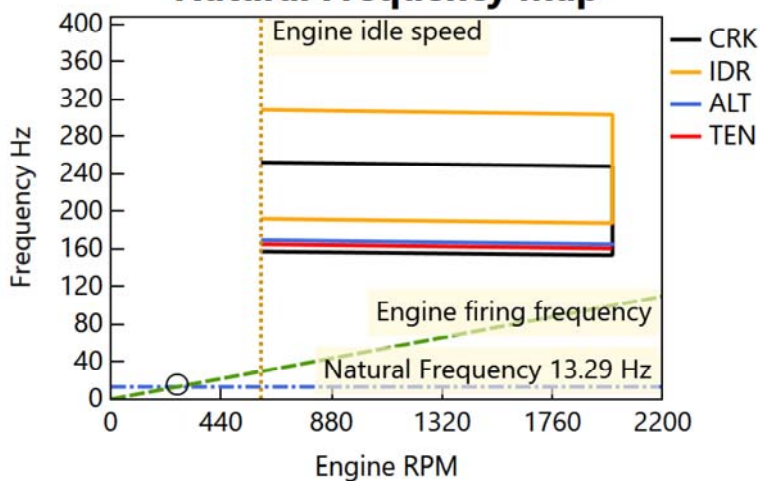
Accessory Data

	File Name - if used	Moment of Inertia kg m ²
IDR	IDR.cmp	0.0002
ALT	AG810-250Amp-ALT.cmp	0.0140
TEN	TEN.cmp	0.0004



RESULTS

Natural Frequency Map



Note: Only low frequency spans are shown.
System Resonance (Mode 1: 15.05)

Note: If Engine Idle is near system resonant RPM, then accessory drive system may have excessive vibration.
If system resonance is within a span frequency window, then span vibration may have excessive amplitude.



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Drive Notes

Date	Note
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